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Course: ECE 601
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Project Proposal

Project Description

(i) The project will utilize the CIFAR-100 dataset, which consists of 60,000 32x32 RGB images. Each image contains 3,072 features (1,024 pixels with 3 channels), resulting in a total input of 184,320,000 pixels across the entire dataset. The data is organized into five training batches and one test batch, each containing 10,000 randomly selected images. Each batch of input data is organized in python as a numpy array of size 10000 x 3072 where the first 1024 entries are the red pixels, the second 1024 are the green pixels, and the third 1024 entries are the blue pixels. Each image is also stored in Row-Major order, that is for each row of 3072 data points the first 32 entries are the top row of red pixels the second 32 entries are the second row of red pixels and so on [1].

The data will be gathered via the torchvision library in Python (using Pytorch will allow me to leverage colabs access to Google's GPU acceleration) but can be directly found and downloaded at;
URL: https://www.cs.toronto.edu/~kriz/cifar.html

(ii) This is a supervised multi-task learning problem. This is because in CIFAR-100 each image comes with two target variables:

1. 100 “fine” labels indicating distinct classes (bee, beetle, etc.).
2. 20 “coarse” labels indicating superclasses (insects, fish, etc.).
3. 2 labels add an additional 120,000 data points.

My primary goal is to build a Convolutional Neural Network (CNN) that can simultaneously infer both labels. Once a baseline performance is established if it is not outside the scope of the class I would like to also focus on model compression via iterative pruning and/or another method. The objective will be to reduce the model footprint by a significant amount while still retaining a very high classification accuracy. I will execute all methods of compression without the help of AI coding tools.

AI Coding Assistant Prompt

Generate a Jupyter Notebook that uses PyTorch to implement a CNN for the CIFAR-100 dataset. The model must handle multi-task classification for both the 100 'fine' classes and the 20 'coarse' superclasses. Train the model and provide a performance evaluation.

References:

[1] A. Krizhevsky, "Learning multiple layers of features from tiny images," University of Toronto, Toronto, ON, Canada, Tech. Rep., 2009. [Online]. Available: https://www.cs.toronto.edu/~kriz/learning-features-2009-TR.pdf